Day 7 Task — Linux Terminal Lab (20 tasks)

Time needed	Deadline	Where to submit
60 minutes	Before the Day 8 session	Portal → Day 7 → Upload

Work inside your **Ubuntu VM** (VirtualBox). First take a snapshot (Machine → Take Snapshot) so you can break things freely. For each task, write the exact command(s) you used and a one-line result. **man <command>**, **--help** and Google are all allowed — real admins use them daily.

Safety rule: never run a command you don't understand, and never run anything with **rm -rf** outside your own ~/seclab folder.

Step 1 — The 20 tasks

#	Area	Task	Command(s) you used	Result / note
1	Navigation	Print your current directory, then list ALL files in your home folder including hidden ones, in long format.		
2	Navigation	Go to /var/log and list the 5 most recently modified files there.		
3	Files	Create a folder ~/seclab with two sub-folders: notes and scripts (one command).		
4	Files	Inside notes, create a file day7.txt containing the line: Linux lab started. Show its contents.		
5	Files	Copy day7.txt to backup.txt, then rename backup.txt to day7-backup.txt.		
6	Search	Find every file under /etc whose name ends in .conf and count them (hint: pipe to wc -l).		
7	Search	Search /etc/passwd for lines containing ‘bash’. How many real login users does your VM have?		
8	Users	Show your username and your groups (two commands). Are you in the sudo group?		
9	Users	Create a new user called analyst with a home directory. (You need sudo.)		

#	Area	Task	Command(s) you used	Result / note
10	Users	Set a password for analyst, then switch to that user with su - analyst. Run whoami, then exit back.		
11	Permissions	Make day7.txt readable and writable by you only (numeric mode). Prove it with ls -l.		
12	Permissions	Create scripts/hello.sh containing echo Hello from $(whoami); make it executable; run it as ./hello.sh		
13	Permissions	Find all SUID binaries on the system. Record the total count and name 3 you recognise (passwd, sudo…).		
14	Processes	List all running processes and find the PID of your shell (hint: ps aux | grep bash, or echo $$).		
15	Processes	Run sleep 300 & in the background, find its PID, then kill it. Confirm it is gone.		
16	Logs	Show the last 20 lines of /var/log/auth.log (or run journalctl -n 20 if the file doesn't exist). What do you see?		
17	Logs	Count how many times the word ‘sudo’ appears in /var/log/auth.log (or journalctl | grep -c sudo).		
18	Network	List every listening TCP/UDP port and the process behind it. Record the ports you see.		
19	Packages	Refresh the package list and install the tool ‘tree’. Run tree ~/seclab to show your folder structure.		
20	Scripting	Write scripts/check.sh that prints the date, your username, and the number of logged-in users (who | wc -l). Make it executable and run it.		

Step 2 — Investigate a suspicious server (thinking exercise)

You are handed a Linux web server that 'feels slow'. Write the six commands you would run, in order, following the Who → What → Where → When → How → Fix routine from the lecture, and say what you'd look for in each.

Step	Command	What would worry you in the output?
Who		
What		
Where		
When		
How		
Fix		

Step 3 — Get ready for Day 8

In the VM run: **sudo apt install wireshark** → choose **Yes** when asked whether non-superusers should capture packets → then **sudo usermod -aG wireshark $USER** and reboot the VM. Tick when done: [ ]

How to submit

- One file: Google Doc, PDF, or clear photos/screenshots of this sheet filled in
- Screenshots of your terminal are welcome (and faster than typing commands twice)
- Must contain: the 20 tasks with commands, the Step 2 table, Step 3 tick
- Upload under Day 7 on the portal before the Day 8 session

Marks (out of 10)

Part	Marks
Tasks 1–20 completed with correct commands (¼ mark each)	5
Step 2 investigation table — sensible commands in the right order	4
Wireshark installed for Day 8	1

Counts toward your lab journal (30%). Partial credit for a wrong command with a right idea. Tomorrow you'll watch these same commands cross the network in Wireshark.